

Ques-11

Sol. Convex lens

Sol. Negative as the image is real and inverted

Ans

Sol. $u = -20$
 $f = 20$

Now,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{(-20)}$$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{20}$$

$$v = 420 \text{ cm}$$

Ques-12

Sol. a)

Refractive index is defined as the ratio of the speed of light in a vacuum to its speed in a specific medium.

Mathematically, refractive index (n) =

$$\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} = \frac{c}{v}$$

Refractive index is a unitless quantity. In terms of angle of incidence (i) and angle of refraction (r).

$$n = \frac{\sin i}{\sin r}$$

Sol. b)

Given, $n_w = \frac{4}{3}$

$$n_g = \frac{3}{2}$$

Speed of light in the glass = $2 \times 10^8 \text{ m/s}$

Now,

$$\text{light in glass } n_g = \frac{c}{v_g} \Rightarrow c = 2 \times 10^8 \times \frac{3}{2}$$

$$c = 3 \times 10^8 \text{ m/s}$$

Now,

$$\text{light in water } n_w = \frac{c}{v_w}$$

$$v_w = \frac{3 \times 10^8 \times 2}{3}$$

$$v_w = 2.25 \times 10^8 \text{ m/s}$$

Ques 8

Sol:

Given,

$$f = +15 \text{ cm}$$

$$u = -10 \text{ cm}$$

Mirror type = Convex

Now,

Finding the image distance from mirror formula.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{16} + \frac{1}{10}$$

$$\frac{1}{f} = \frac{50}{30} + \frac{5}{30}$$

$$f = \frac{50+5}{30}$$

$$f = +6 \text{ cm}$$

Ques 9

Sol:

Given,

$$u = -10 \text{ cm}$$

$$m = 3$$

We know,

$$m = -\frac{v}{u}$$

$$m = \frac{v}{10}$$

The image is formed 6 cm behind the mirror, it is a virtual and erect image.

Ques 10

Sol:

$$u = \text{infinity}$$

$$v = -80 \text{ cm}$$

$$f = ?$$

Now, 3 cm magnification concave lens is used to correct the problem.

So, focal length = Distance of far point from eye

Hence,

$$f = -80 \text{ cm} = -0.8 \text{ m}$$

$$\text{Power of lens} = \frac{1}{f} = \frac{1}{-0.8} = -1.25 \text{ D}$$

So, the image is real and it is formed

$$f = 7.1 \text{ cm}$$

$$f = \frac{30}{4}$$

$$\frac{1}{f} = \frac{4}{30}$$

$$\frac{1}{f} = \frac{1}{30} - \frac{1}{10}$$

$$\text{focal length} = \frac{1}{\frac{1}{f} = \frac{1}{30} - \frac{1}{10}}$$

$$v = 30 \text{ cm}$$

PHYSICS

Ques-1
Sol: (3) -1

Ques-2
Sol: (2) outer surface of convex

Ques-3
Sol: (4) Virtual, erect, enlarged

Ques-4
Sol: (1) Both A and R is correct and R is the correct explanation of A.

Ques-5
Sol: Both is true but R is not the correct explanation of A

Ques-6
Sol: The two common defects of vision are -

- i) Myopia can be corrected by using concave lens
- ii) Hypermetropia can be corrected by using a convex lens.

Ques-7
Sol: Rays of the light from the tip of the pencil bend away from the normal as they go from

Ques-8
Sol: water to air. The refracted rays appear to come from point. Thus, the immersed portion in water appears bent and short.

Ques-9
Sol: b)

Now,
 $m = -2$
 $v = +20 \text{ cm}$
 $f = ?$

$m = \frac{v}{u}$
 $-2 = \frac{20}{u}$

$-u = \frac{20}{2}$
 $-u = 10$

$u = -10$

Now,

by using lens formula

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{20} - \frac{1}{(-10)} = \frac{1}{f}$

$\frac{1}{20} + \frac{1}{10} = \frac{1}{f}$

$\frac{3}{20} = \frac{1}{f}$

$\frac{20}{3} = f$

$6.67 \text{ cm} = f$